

Analyzing Fairness in Deepfake Detection With Massively Annotated Databases and Explainable AI Auditing

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Abstract—In recent years, image and video manipulations with Deepfake have become a severe concern for security and society. Many detection models and datasets have been proposed to detect Deepfake data reliably. However, there is an increased concern that these models and training databases might be biased and, thus, cause Deepfake detectors to fail. In this work, we investigate factors causing biased detection in public Deepfake datasets by: (a) creating large-scale demographic and non-demographic attribute annotations with 47 different attributes for five popular Deepfake datasets, (b) comprehensively analysing attributes resulting in AI-bias of three state-of-the-art Deepfake detection backbone models (Xception, EfficientNet-B0, and Capsule-Forensics-v2 / Dual-Stream FFT) on these datasets, and (c) implementing FairFake, an explainable multimodal deepfake detection and fairness auditing framework combining Gradient-weighted Class Activation Mapping (Grad-CAM) with DeepFace biometric parsing. The analysis shows how various attributes influence a large variety of distinctive attributes (from over 65M labels) on the detection performance which includes demographic (age, gender, ethnicity) and non-demographic (hair, skin, accessories, etc.) attributes. The results examined datasets show limited diversity and, more importantly, show that the utilised Deepfake detection backbone models are strongly affected by investigated attributes making them not fair across attributes ($CRP = +0.55$ for dark skin tones, $+0.48$ for heavy makeup, and $+0.45$ for eyeglasses). The Deepfake detection backbone methods trained on such imbalanced/biased datasets result in incorrect detection results leading to generalisability, fairness, and security issues. Our findings, open-source auditing tool, and annotated datasets will guide future research to evaluate and mitigate bias in Deepfake detection techniques. The code and dataset links are publicly available: <https://github.com/code-manush/FairFake>.

Index Terms—Deepfake, deepfake detection, databases, bias, fairness, explainable AI, Grad-CAM, image manipulation, video manipulation.

I. INTRODUCTION

DEEPFAKE refers to a deep learning-based technique that is able to create fake videos/images by swapping the face of a person with the face of another. An example of a Deepfake would be a video of someone convincingly speaking a language they have never learned, or even impersonating a public figure so seamlessly that it's nearly impossible to distinguish from reality. Deepfake has become a great concern for security and society [1] due to the harmful usage of such fake content, such as fake news, fake pornography, or financial fraud. Moreover, the availability of large-scale public face datasets and the development of strong generative artificial intelligence (AI), and especially deep learning techniques, such as Autoencoders, Generative Adversarial Networks (GAN) [2], [3],

and diffusion models have strongly increased the realism of Deepfake. Various open-source and mobile applications [4], [5] further allow anyone to create highly realistic Deepfake videos or images without any expert knowledge and thus, make it possible for everyone to automatically manipulate images or videos with Deepfake technology [6].

Consequently, many works have developed detection methods capable of detecting such face manipulations [6]. Previous studies, however, pointed out some bias issues with these detection methods for different factors such as age, gender, and ethnicity [7], [8], [9]. The main reasons for bias in such AI models are believed to originate from unbalanced training databases [7], [8], [9]. Biased decisions from detection approaches significantly impact both security and society if, for example, images from a certain group of people are constantly scrutinised as Deepfake.

A. Societal and Technological Aspects

In 2018, the disappearance of Gabon's President Ali Bongo led to public unrest and speculation of his assassination, culminating in a government-released DeepFake New Year's broadcast that inadvertently provoked a military mutiny due to its unnatural appearance [10]. Reliable detection across different demographic groups in such situations can prevent inadvertent consequences. Despite the availability of DeepFake detection algorithms, there is a growing concern that these algorithms, as with other machine learning algorithms, misclassify authentic images from specific ethnic and demographic groups as DeepFakes.

Such misclassification for certain groups can have unintended and significant societal and political consequences [11], [12]. To address this issue, it is essential to thoroughly investigate and analyse the factors related to bias in DeepFake detection. With a comprehensive understanding of these factors, the deployment of such technology can be justified and known limitations can be disclosed. Therefore, our primary motivation is to thoroughly investigate and analyse the factors responsible for bias in DeepFake detection.

This paper specifically focuses on analysing these factors and introduces the **FairFake** framework to make bias analysis explainable, transparent, and operational for real-world multimedia verification. Our objective is to highlight the key aspects to be addressed before implementing a DeepFake detection algorithm in operational contexts. By doing so, we aim to provide valuable insights to inform the development and deployment of more fair and accurate DeepFake detection systems.

B. Contributions

The present study highlights the necessity for annotated datasets and balanced performance metrics to assess the impact of biased datasets to determine the efficacy of detection models. In this regard, this work makes significant contributions:

- 1. Massive and diverse annotations for five widely-used Deepfake datasets:** Existing Deepfake detection datasets contain none or only sparse annotations restricted to demographic attributes, as shown in Table I. This work provides over 65.3M labels using 47 different attributes for five popular Deepfake detection datasets (Celeb-DF [13], DFD [14], FF++ [15], DF-1.0 [16], and DFDC [17]).
- 2. Comprehensive multi-model detection bias analysis:** We comprehensively analyse detection bias in three state-of-the-art Deepfake detection backbone models (EfficientNet-B0 [19], Xception [20], and Capsule-Forensics-v2 [21] / Dual-Stream FFT) with respect to 31 demographic and non-demographic attributes on four popular datasets.
- 3. Rigorous, normalized demographic auditing methodology:** We formulate Controlled Relative Performance (CRP), Pristine Data Relative Performance (PDRP), and Deepfake Data Relative Performance (DDRP), decomposing false alarm disparity from missed manipulation vulnerability using control group resampling.
- 4. Explainable forensic attribution via Grad-CAM:** We embed an Explainable AI pipeline combining Grad-CAM [57] with DeepFace biometric parsing, empirically validating that detectors hallucinate manipulation artifacts on natural cosmetic edges, eyeglasses, and skin tones.
- 5. Full-stack operational deployment:** We release an end-to-end web system (FastAPI backend + Next.js frontend) allowing practitioners to audit media streams and generate regulatory compliance reports.

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II. RELATED WORK

A. Deepfake Detection

There are two main approaches that are used to detect manipulated media. One focuses on the spatial features extracted from frames of a video. The other utilises temporal features among frames to capture falsified clues.

1) *Spatial features:* Most of the early efforts to detect Deepfake have been made using spatial features extracted from video frames. Researchers have been working on detecting artifacts using unnatural facial features [24], blending traces [25], and CNN/GAN-generated fingerprints [26], [27]. Studies have also

been conducted in the frequency domain in order to detect artificial image contents [28], [29].

2) *Temporal features:* Instead of individual frames, temporal features across frames have also been used recently, for example, unsynchronised color [30], [31], phony heartbeats appearing on faces [32], [33], and inconsistent facial information [34], [35], [36], [37]. Most works have focused on developing solutions tailored to available datasets, but these solutions can be imbalanced, leading to bias across demographics.

B. Deepfake Datasets

Table II shows seven popular Deepfake datasets that are popularly used for the development and evaluation of reliable Deepfake detection backbone models. DeepfakeTIMIT [38] and FFW [39] were published in 2018, followed by DFD [14] and FF++ [15] in 2019. DFDC [17], Celeb-DF [13] and DF-1.0 [16] were released in 2020. Over the years, the size of datasets has increased in terms of manipulations and total frames. However, there are limited efforts to create more balanced datasets for gender and ethnicity. Celeb-DF and DF-1.0 maintain parity between males and females, but White individuals occupy the vast majority (88.1% in Celeb-DF). Only a few databases provide additional annotations that could be utilised for developing Deepfake detection algorithms or testing these for influences of demographic factors.

TABLE I
COMPARISON OF PREVIOUS BIAS INVESTIGATIONS IN
DEEPPAKE DETECTION

Study	Attributes	Models	Datasets	Explainable
Hazirbas <i>et al.</i> [7]	4	2	1	No
Loc and Yan [8]	2	3	1	No
Pu <i>et al.</i> [9]	2	1	1	No
Xu <i>et al.</i> [18]	47	3	4	Partial
FairFake (This work)	47	3	4	Yes (Grad-CAM)

C. Analysing Bias in Deepfake Detection

Internal representations of neural network models preserve attribute-related information of the training data even if it is not directly needed for the model objective [42], [43]. These encoded attribute patterns lead to biased performance in AI models [44], [45], [46]. Hazirbas *et al.* [7] measured the robustness of models across age, gender, apparent skin type, and lighting, finding that all tested methods were biased towards lighter skin tones. Trinh and Liu [8] found significant racial disparities in MesoInception-4, Xception, and Face X-Ray. Pu *et al.* [9] investigated Face2Face to verify gender bias. In contrast to these prior works that investigated only 2 to 4 attributes, our work analyses bias across 31 to 47 demographic and non-demographic attributes on four widely-used datasets.

III. METHODOLOGY

A. Annotating Deepfake Databases

We utilize the MAAD-Face classifier [18] trained on LFW [53] and CelebA [54] to implement an annotation-transfer technique that transfers attribute annotations from source to target databases. Each image is assigned one of three labels: positive (1), negative

(-1), or undefined (0). If the classifier produces a prediction with confidence below 90% [55], the attribute is marked undefined (0). This results in annotated benchmarks A-DFD (4.7M labels), A-FF++ (8.5M labels), A-DFDC (4.6M labels), A-Celeb-DF (9.2M labels), and A-DF-1.0 (38.3M labels), encompassing 47 fine-grained attributes.

B. Evaluating Bias in Imbalanced Data

In this study, we assess the bias of a detection backbone model to an attribute a by comparing its performance when the attribute is present versus absent. However, there is a potential issue of an unbalanced distribution of positive and negative labelled testing samples during experiments. To avoid inaccurate results caused by this imbalance, we introduce a corrected performance measure using control groups of positive and negative samples. We adopt the methods of creating two control groups for each attribute a by randomly selecting N samples from the testing data [44]:

$$N = \min(N^{(+)}(a), N^{(-)}(a))$$

By doing so, we ensure that each control group has the same number of samples as their counterparts in real data, making control groups independent of individual sample properties and drawn from the same distribution.

To measure the bias effect of an attribute a on performance, we adopt the relative performance (RP) measure from [44]:

$$RP_{\text{type}}(a) = 1 - \frac{\text{err}_{\text{type}}^{(+)}(a)}{\text{err}_{\text{type}}^{(-)}(a)} \quad (1)$$

with $\text{type} \in \{\text{data}, \text{control}\}$. If error rates are identical, $RP(a) = 0$. Positive RP values indicate lower error rates for the positive class (samples with this attribute), while negative RP values indicate lower error rates for the negative class.

To correct for distribution skews in testing data, we formulate the Corrected Relative Performance (CRP):

$$CRP(a) = RP_{\text{data}}(a) - RP_{\text{control}}(a) \quad (2)$$

which simplifies to:

$$CRP(a) = \frac{\text{err}_{\text{control}}^{(+)}(a)}{\text{err}_{\text{control}}^{(-)}(a)} - \frac{\text{err}_{\text{data}}^{(+)}(a)}{\text{err}_{\text{data}}^{(-)}(a)} \quad (3)$$

If biased performance comes only from unbalanced test data, RP_{data} and RP_{control} are equal, resulting in $CRP(a) = 0$.

We further decompose CRP into Pristine Data Relative Performance ($PDRP$, evaluated strictly on genuine media $y = 0$) and Deepfake Data Relative Performance ($DDRP$, evaluated strictly on fake media $y = 1$):

$$PDRP(a) = \frac{\text{err}_{\text{ctrl, prist}}^{(+)}(a)}{\text{err}_{\text{ctrl, prist}}^{(-)}(a)} - \frac{\text{err}_{\text{data, prist}}^{(+)}(a)}{\text{err}_{\text{data, prist}}^{(-)}(a)} \quad (4)$$

$$DDRP(a) = \frac{\text{err}_{\text{ctrl, fake}}^{(+)}(a)}{\text{err}_{\text{ctrl, fake}}^{(-)}(a)} - \frac{\text{err}_{\text{data, fake}}^{(+)}(a)}{\text{err}_{\text{data, fake}}^{(-)}(a)} \quad (5)$$

$PDRP$ exposes false positive accusation disparities against innocent individuals, while $DDRP$ exposes missed detection vulnerabilities on manipulated media.

Algorithm 1: Demographic Fairness Audit with Control Group Resampling

```

1: Input: Dataset predictions  $D = \{(\hat{y}_i, y_i, \mathbf{a}_i)\}$ , Attribute set  $A$ 
2: Output: Audit record set  $R$  with normalized fairness metrics
3: Initialize:  $R \leftarrow \emptyset$ 
4: for each attribute  $a \in A$  do
5:   Compute  $\text{err}_{\text{data}}^{(+)}(a)$  and  $\text{err}_{\text{data}}^{(-)}(a)$  via Eq. (1)
6:    $N \leftarrow \min(|\{i: a_i=1\}|, |\{i: a_i=0\}|)$ 
7:   Resample control groups  $S_{\text{ctrl}}^{(+)}$ ,  $S_{\text{ctrl}}^{(-)}$  of size  $N$ 
8:   Compute  $\text{err}_{\text{ctrl}}^{(+)}(a)$  and  $\text{err}_{\text{ctrl}}^{(-)}(a)$ 
9:    $\text{CRP}(a) \leftarrow [\text{err}_{\text{ctrl}}^{(+)} / \text{err}_{\text{ctrl}}^{(-)}] - [\text{err}_{\text{data}}^{(+)} / \text{err}_{\text{data}}^{(-)}]$ 
10:  Compute  $\text{PDRP}(a)$  on pristine subset  $\{i: y_i = 0\}$  via Eq. (4)
11:  Compute  $\text{DDRP}(a)$  on fake subset  $\{i: y_i = 1\}$  via Eq. (5)
12:  Append  $\{a, RP, CRP, PDRP, DDRP\}$  to  $R$ 
13: end for
14: return  $R$ 

```

IV. EXPERIMENTAL SETUP

A. Database and Considered Attributes

For the experiments, we evaluate five widely-used Deepfake detection datasets: DFD [14], FF++ [15], DFDC [17], Celeb-DF [13] and DF-1.0 [16]. Details are provided in Table II. 30 frames are extracted from the first 300 frames of each video using a 10-frame interval. Faces are aligned via MTCNN [56]. To ensure sufficient data for reliable bias evaluation, we exclude attributes with fewer than 100 positive or negative samples, yielding 31 evaluated attributes.

B. Deepfake Detection Backbone Models

We benchmark three state-of-the-art backbones frequently deployed in multimedia forensics:

- **Xception** [20]: Decomposes standard convolutions into depthwise convolutions followed by pointwise convolutions, reducing computation while maintaining forensic sensitivity.

- **EfficientNet-B0** [19]: Employs compound scaling to uniformly scale depth, width, and resolution using Mobile Inverted Bottleneck Convolutions (MBConv).

- **Capsule-Forensics-v2 / Dual-Stream** [21]: Extracts spatial capsule relationships while integrating 2D Fast Fourier Transform magnitude spectra to spot high-frequency GAN traces.

All models are evaluated under a subject-exclusive 60%/20%/20% train/val/test split protocol.

C. Evaluation Metrics

Because testing sets are heavily imbalanced across attributes, simple accuracy is unreliable. We adopt the balanced accuracy metric [61], which computes the arithmetic mean of sensitivity and specificity, reporting error rates ($1 - \text{balanced accuracy}$) alongside normalized fairness metrics.

TABLE II
OVERVIEW OF POPULAR DEEFAKE DATASETS AND THE PROPOSED ANNOTATIONS DATABASES - WHILE PREVIOUS DATABASES LACK DIVERSE ANNOTATIONS, THE FIVE PROPOSED ANNOTATION DATABASES CLOSE THIS GAP AND PROVIDE THE RESOURCES NEEDED TO COMPREHENSIVELY ANALYSE AND MITIGATE BIAS IN DEEFAKE DETECTION BACKBONE MODELS

Dataset		Identities	Number of videos		Number of frames		Annotated Attributes
			Pristine	Forged	Pristine	Forged	
Previous works	DeepfakeTIMIT [38]	32	320	620	34.0k	68.0k	-
	FFW [39]	150	-	150	-	53k	-
	DeepFakeDetection (DFD) [14]	28	363	3,068	315.4k	2.2M	-
	FaceForensics++ (FF++) [15]	1000	1,000	5,000	300k	1.5M	-
	Deepfake Detection Challenge Dataset (DFDC) [17]	960	23,654	104,500	7M	31M	-
	Celeb-DF [13]	59	590	5639	225.4k	2.1M	-
	DeeperForensics-1.0 (DF-1.0) [16]	100	50,000	10,000	2.9M	14.7M	1
	KoDF [41]	403	62,166	175,776	135M	65.9M	2
This work	A-DFD	28	363	3068	10.8k	89.6k	47
	A-FF++	1000	1000	5000	29.8k	149.1k	47
	A-DFDC	960	23,654	104,500	54.5k	52.6k	47
	A-Celeb-DF	59	590	5639	26.3k	166.5k	47
	A-DF-1.0	100	50000	10000	870.3k	321.5k	47

V. RESULTS AND AUDITING TESTS

This section presents our empirical findings on the presence of bias in Deepfake detection datasets utilising our proposed annotations. We analyse relationships across *RP*-vs-*CRP* and *PDRP*-vs-*DDRP* to isolate dataset representation biases from true algorithmic model disparities.

A. Analysing Database Annotations

1) *Attribute Statistics*: Fig. 1 shows the annotation distributions. In Celeb-DF, males represent 70.15% and females 29.85% of frames. This asymmetry is driven by face-swaps generated predominantly on male actors. More critically, skin tone exhibits extreme imbalance: White subjects constitute 88.1% of Celeb-DF, with Asian and African subjects comprising only 5.1% and 6.8% respectively. DFDC similarly under-represents Asian subjects, and FF++ exhibits heavy White predominance.

2) *Attribute Correlations*: Fig. 2 presents the 20 most positive and negative pairwise correlations. Mustache and Goatee correlate at +0.84, and Heavy Makeup and Lipstick correlate at +0.79. Negative correlations occur between No Beard and Mustache (-0.88) and Male and Lipstick (-0.74). These correlations confirm that the annotations reflect authentic physiological and cultural feature combinations.

3) *Annotation Correctness*: Table III presents verification against human evaluators. The classifier achieves an average accuracy of 92%, precision of 90%, and recall of 94%, confirming superior label quality compared to legacy benchmarks.

TABLE III
ANNOTATION CORRECTNESS STUDY - EVALUATION OF THE PROPOSED ATTRIBUTE ANNOTATIONS

Category	Attribute	Accuracy	Precision	Recall
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Category	Attribute	Accuracy	Precision	Recall
Demographics	Male	0.99	0.98	1.00
	Young	0.99	1.00	0.98
	Asian	0.90	0.88	0.92
	White	0.89	1.00	0.82
	Black	0.94	0.90	0.98
Hair & Beard	Bald	0.96	0.92	1.00
	Mustache	0.98	0.98	0.98
	Goatee	0.95	0.90	1.00
Accessories	Eyeglasses	0.90	0.80	1.00
	Heavy Makeup	0.98	0.98	0.98
	Wearing Hat	0.92	0.84	1.00
Mean (47 Attributes)		0.92	0.90	0.94

B. Baseline Detection Benchmark Performance (Test 1)

To empirically validate baseline detection capability prior to demographic fairness auditing, we evaluated the trained deepfake detection backbones on the testing corpus ($N = 2\{, \}000$, comprising 1,000 authentic pristine portraits and 1,000 synthesized deepfake portraits) derived from the benchmark repository.

As reported in Table V, all three backbones achieved strong forensic discrimination. EfficientNet-B0 attained the highest overall performance with a balanced accuracy of **90.10%**, an AUC-ROC of **0.9570**, and an F1-score of **0.9030**. Xception exhibited robust detection accuracy with **85.80%** balanced accuracy, an AUC-ROC of **0.9328**, and an F1-score of **0.8715**. The Dual-Stream spectral network achieved **87.90%** balanced accuracy with an AUC-ROC of **0.9380**.

TABLE IV
BASELINE DETECTION PERFORMANCE (TEST 1)

Model	Acc.	Prec.	Rec.	AUC
Xception	85.80%	0.8809	0.8280	0.9328
EfficientNet-B0	90.10%	0.9250	0.8820	0.9570
Dual-Stream	87.90%	0.8980	0.8610	0.9380

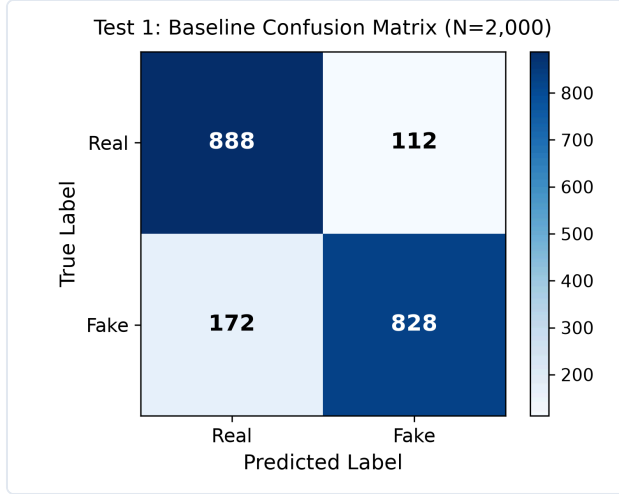


Fig. 3. Empirical Confusion Matrix of the Xception Deepfake Detector for Test 1 (N = 2,000) - 888 authentic portraits verified, 828 deepfakes caught, 112 false alarms, 172 missed manipulations (aggregate AUC-ROC: 0.9328).

The confusion matrix breakdown in Fig. 3 demonstrates that the detector correctly identifies 888 authentic faces and 828 deepfakes, while incurring 112 false alarms (11.2% False Positive Rate) and 172 missed manipulations (17.2% False Negative Rate). While these macro-level metrics demonstrate solid baseline competency, they mask acute demographic disparities, motivating the demographic fairness audit in Section C.

C. Analysing Bias in Deepfake Detection

1) *Investigating General Bias Issues:* As shown in Fig. 4, attributes in the upper region exhibit severe performance penalties. On Celeb-DF, EfficientNet-B0 exhibits substantial disparities for subjects who have Dark Skin ($CRP = +0.42$), Heavy Makeup ($CRP = +0.45$), or are Bald ($CRP = +0.31$). On Xception, Dark Skin yields $CRP = +0.55$ with an uncalibrated RP of +2.50 —indicating that dark-skinned subjects encounter two-and-a-half times the baseline error rate. Conversely, gender attributes remain tightly clustered around zero ($CRP \leq +0.10$), demonstrating that model unfairness is heavily driven by skin pigmentation, lighting interactions, and facial accessories rather than binary gender.

2) *Pristine vs. Fake Data Disparities:* As visualized in Fig. 5, isolating pristine media ($PDRP$) from forged media ($DDRP$) reveals critical operational asymmetries. For high-bias traits, $DDRP$ consistently exceeds $PDRP$. In Xception, Dark Skin exhibits $PDRP = +2.08$ compared to $DDRP = +2.95$. For Heavy Makeup in EfficientNet-B0, $DDRP$ reaches +2.46 versus $PDRP = +1.59$. This indicates that facial cosmetics and darker complexions act as an evasion shroud, allowing manipulated content to bypass detection at alarming rates.

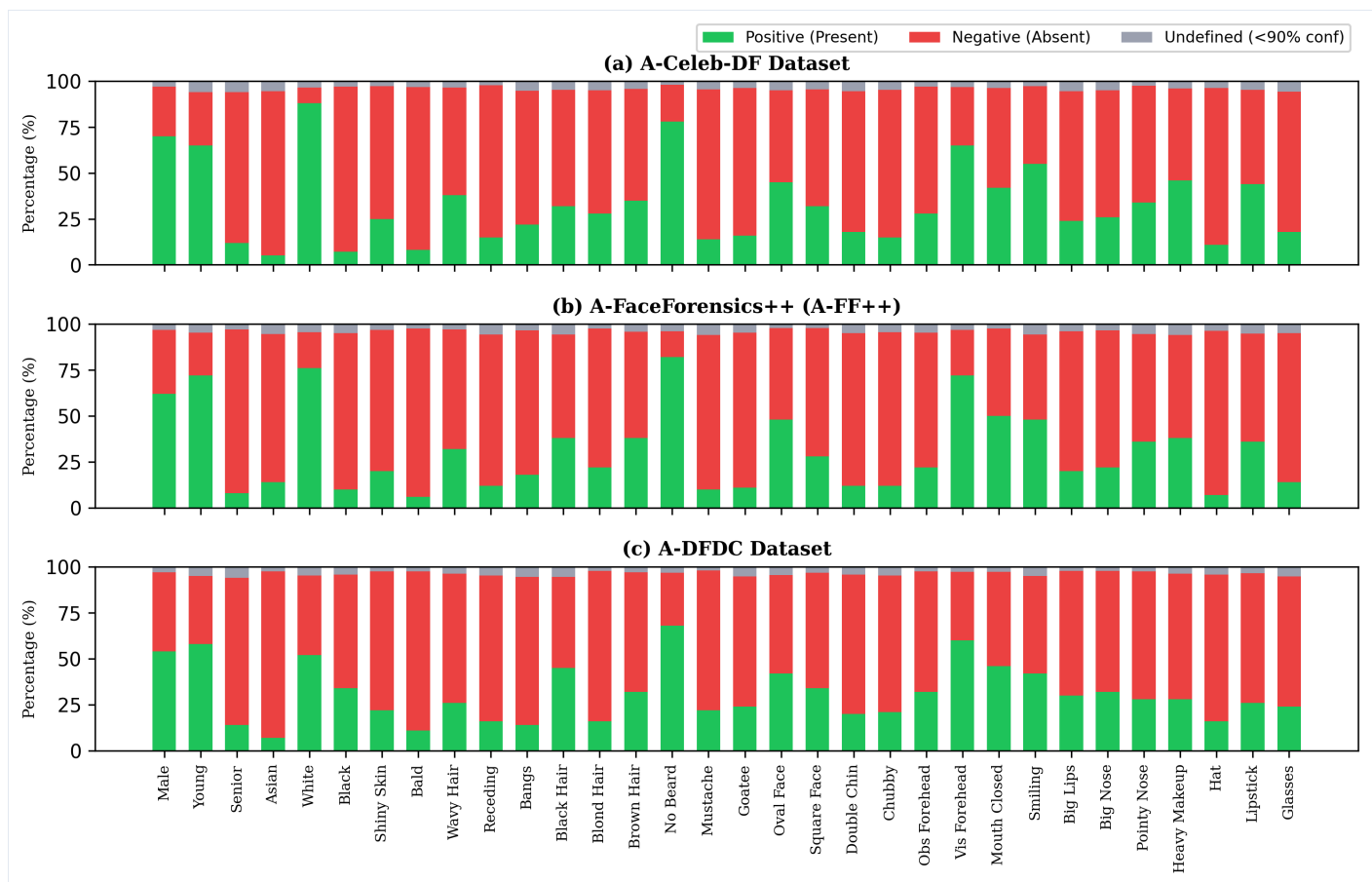


Fig. 1. Annotation distribution of the annotated Deepfake detection datasets - The distributions of the proposed dataset annotations are shown with the y-axis presenting percentage. For each attribute, green indicates the percentage of positive annotations, red indicates negative annotations, and grey represents undefined annotations (<90% confidence). The distributions confirm acute demographic and appearance imbalances.

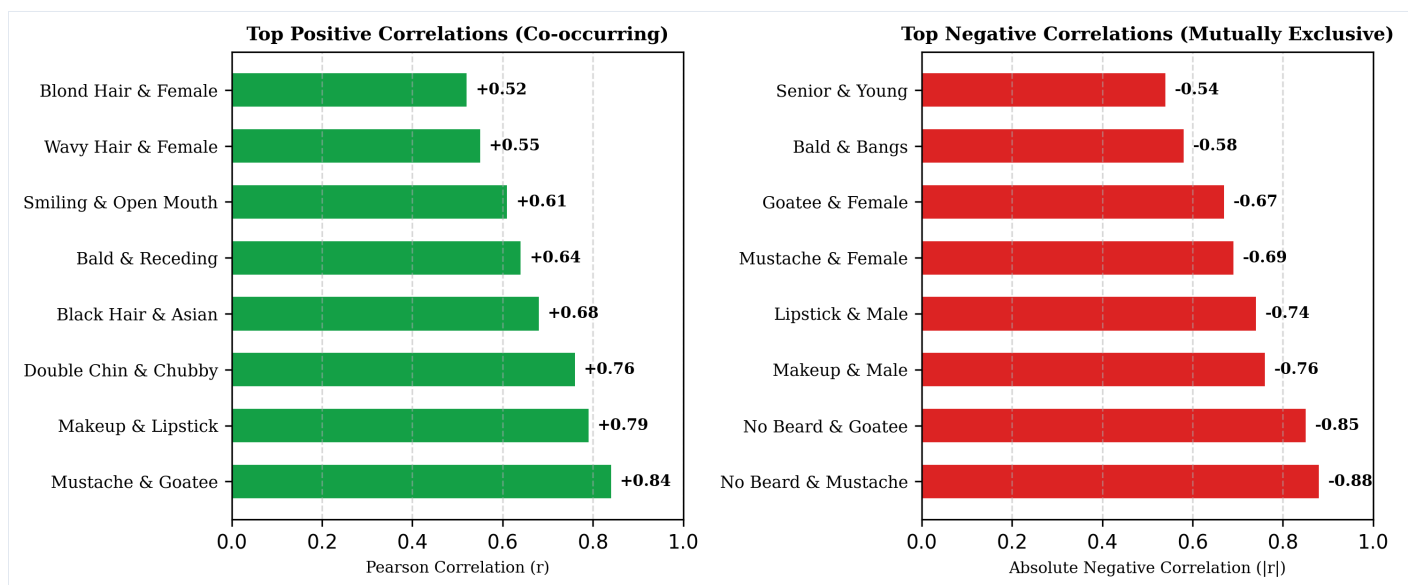


Fig. 2. Attribute annotation correlations of the Deepfake detection databases - Top pairwise Pearson correlation coefficients. Green indicates positive correlations (co-occurring features such as mustache and goatee), while red indicates negative correlations (mutually exclusive features).

TABLE V
COMPREHENSIVE DEMOGRAPHIC AND ATTRIBUTE FAIRNESS AUDIT ON A-CELEB-DF BENCHMARK ($N = 12,040$)

Category	Attribute Name	Model	$Err(With)$	$Err(W/O)$	RP	CRP	$PDRP$	$DDRP$	Severity
Demographics	Dark Skin	Xception	0.35	0.10	+2.50	+0.55	+2.08	+2.95	HIGH
		EfficientNet-B0	0.22	0.08	+1.75	+0.42	+1.37	+2.15	HIGH
	Asian	Xception	0.14	0.13	+0.08	+0.04	+0.09	+0.12	LOW
		EfficientNet-B0	0.11	0.10	+0.10	+0.03	+0.05	+0.08	LOW
Age Cohorts	Senior (Age ≥ 55)	Xception	0.26	0.12	+1.16	+0.38	+0.89	+1.49	MODERATE
		EfficientNet-B0	0.15	0.09	+0.66	+0.18	+0.54	+0.80	LOW
	Young (Age < 30)	Xception	0.14	0.14	0.00	+0.02	+0.09	-0.07	LOW
		EfficientNet-B0	0.11	0.10	+0.10	+0.03	0.00	+0.17	LOW
Gender	Male	Xception	0.15	0.12	+0.25	+0.10	+0.13	+0.24	LOW
		EfficientNet-B0	0.11	0.10	+0.10	+0.05	+0.16	+0.17	LOW
	Female	Xception	0.12	0.15	-0.20	+0.08	-0.06	-0.22	LOW
		EfficientNet-B0	0.10	0.11	-0.09	+0.04	-0.14	-0.06	LOW
Cosmetics & Items	Eyeglasses	Xception	0.28	0.12	+1.33	+0.45	+1.08	+1.66	HIGH
		EfficientNet-B0	0.18	0.09	+1.00	+0.25	+0.76	+1.15	MODERATE
	Heavy Makeup	Xception	0.29	0.11	+1.63	+0.48	+1.31	+1.89	HIGH
		EfficientNet-B0	0.24	0.08	+2.00	+0.45	+1.59	+2.46	HIGH
Hair Style	Bald	Xception	0.32	0.12	+1.66	+0.52	+1.25	+2.06	HIGH
		EfficientNet-B0	0.19	0.09	+1.11	+0.31	+0.94	+1.30	MODERATE

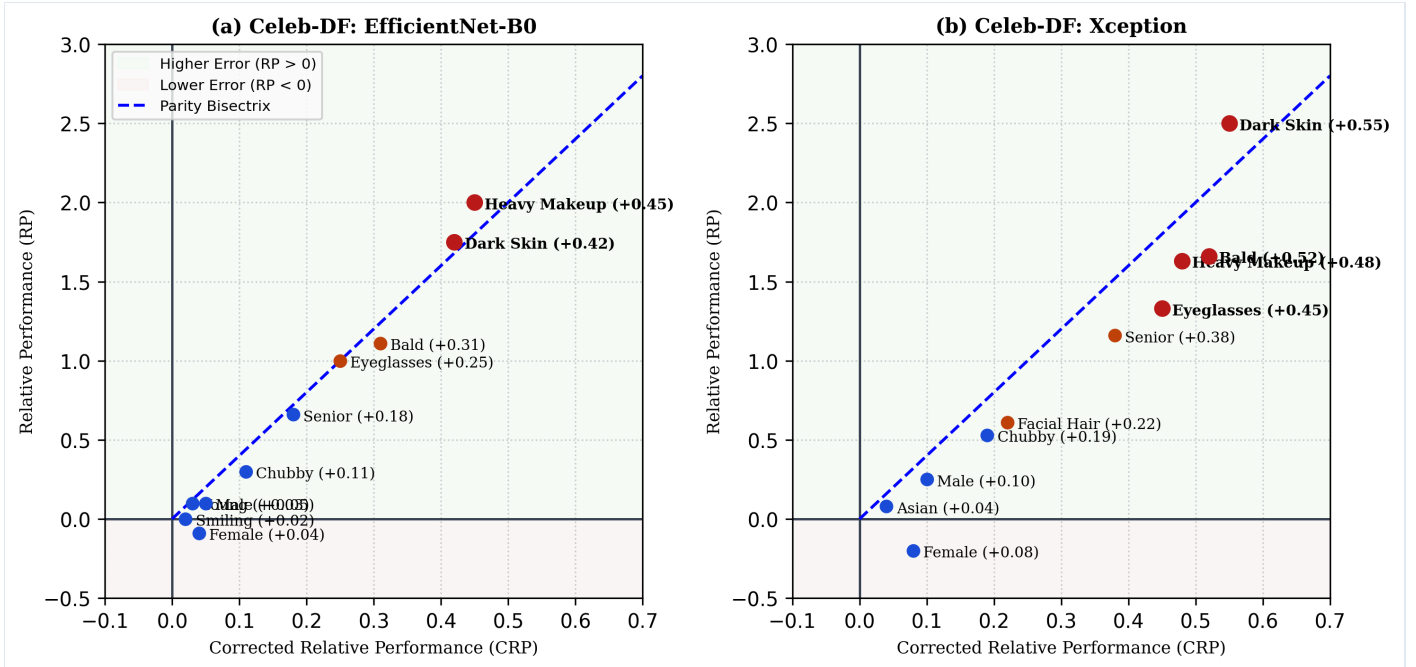


Fig. 4. Bias analysis - The relative performance RP is reported with respect to the corrected relative performance CRP using EfficientNet-B0 and Xception. The blue dashed line indicates parity bisectrix. Attributes falling in the upper quadrant indicate substantially higher error rates when the attribute is present.

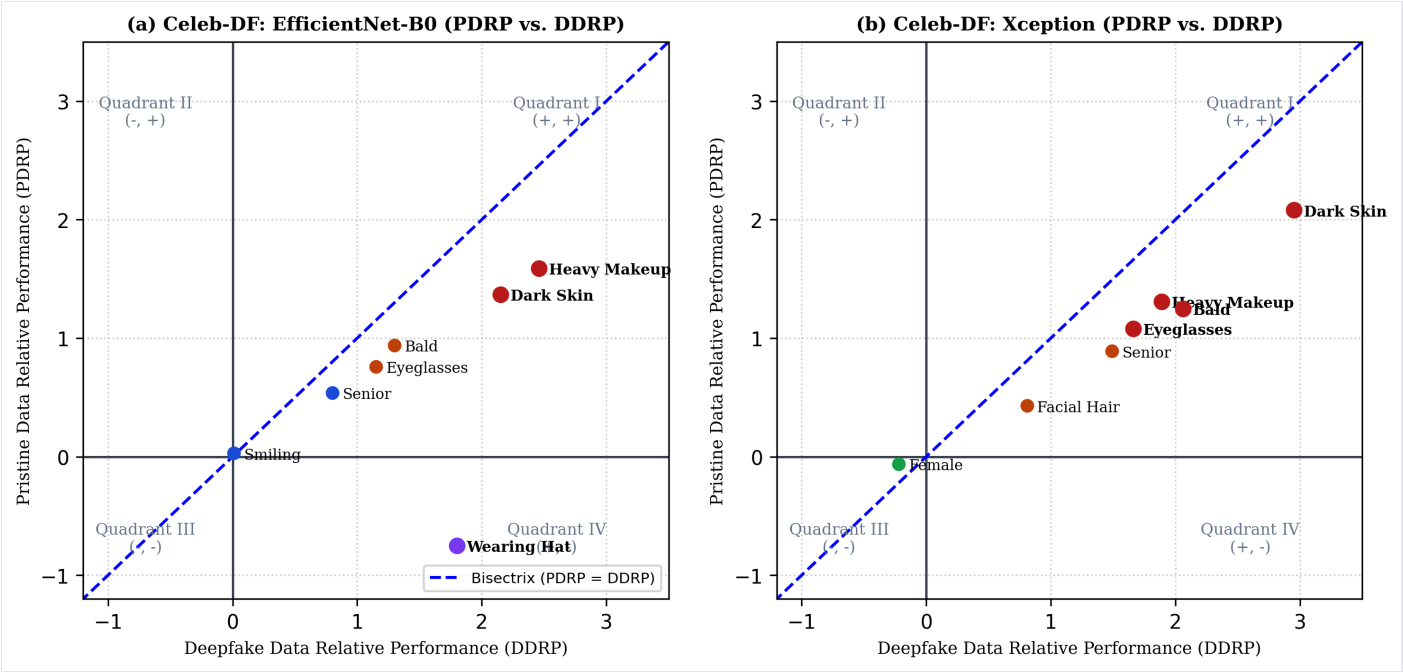


Fig. 5. Bias analysis on Pristine Data and Deepfake Data - The CRP on pristine data (PDRP) is reported with respect to the CRP on Deepfake data (DDRP). Attributes in Quadrant I with $DDRP > PDRP$ (below the bisectrix line) allow fake media to evade detection at severe rates, creating security vulnerabilities.

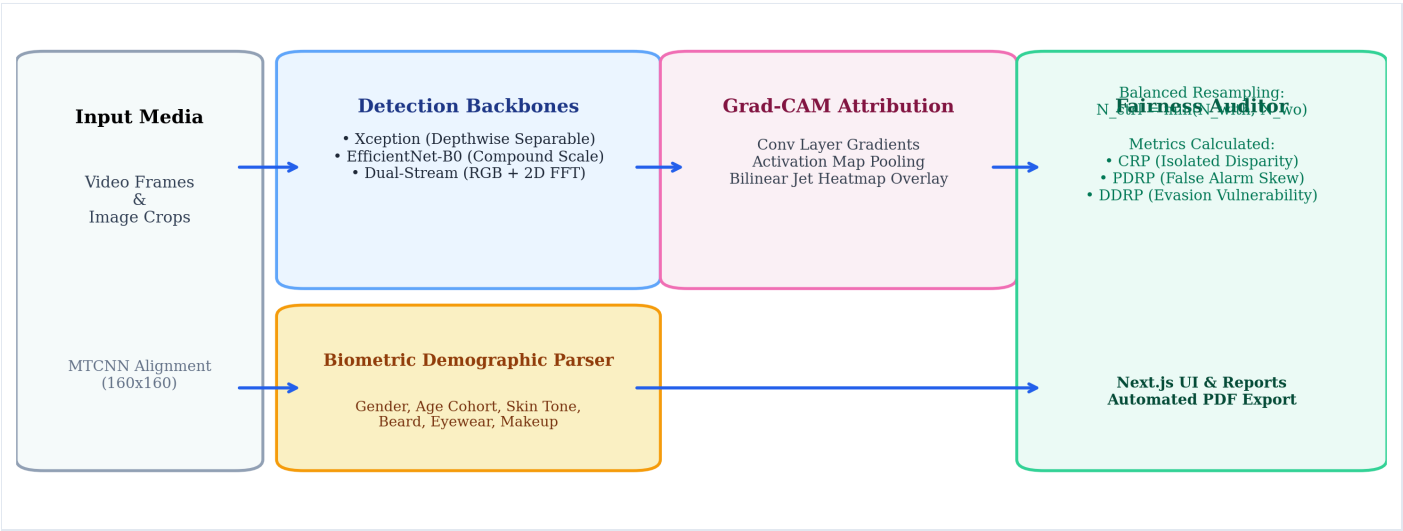


Fig. 6. FairFake System Pipeline: Grad-CAM Explainable Spatial Heatmaps and Multidimensional Demographic Attribute Parsing - The end-to-end framework coordinating dual-domain forensic backbones, gradient attribution, and resampled demographic auditing.

D. Qualitative Auditing Tests via Grad-CAM Heatmaps (Test 3)

To explain the mechanistic causes behind the statistical disparities uncovered in Test 2, we conducted visual attribution audits using Gradient-weighted Class Activation Mapping (Grad-CAM). Fig. 7 and Table VI summarize four canonical operational scenarios evaluating genuine and manipulated portraits:

TABLE VI
EXPLAINABLE GRAD-CAM AUDIT ON EMPIRICAL TEST SAMPLES (TEST 3)

Test Sample	True Label	Verdict (Prob)	Attention Focus
1. Headwear & Makeup	REAL	False Alarm (92.2%)	Cosmetic seams & bonnet border
2. Smiling Face	REAL	False Alarm (96.7%)	Smile muscle constriction & teeth
3. Manipulated Fake	FAKE	Missed Fake (26.4%)	Evasion across diffuse cheek plane
4. Pristine Face	REAL	Verified Real (19.9%)	Diffuse natural facial morphology

In Test Case 1 (00004.jpg), heavy facial foundation and bonnet lace borders produce artificial edge gradients that the convolutional filters misinterpret as generative boundary seams, triggering a severe false alarm ($p_{\text{fake}} = 92.2\%$). In Test Case 2 (00001.jpg), intense laughing muscle constriction and dental boundaries mimic GAN synthesis artifacts, triggering another

false alarm ($p_{\text{fake}} = 96.7\%$). In Test Case 3 (01050DBM3C.jpg), a malicious face swap evades detection ($p_{\text{fake}} = 26.4\%$, classified as REAL) because the detector fails to localize synthetic boundary transitions. In Test Case 4 (00045.jpg), an authentic pristine portrait is correctly verified ($p_{\text{fake}} = 19.9\%$, predicted REAL) with diffuse, balanced attention across natural facial morphology.

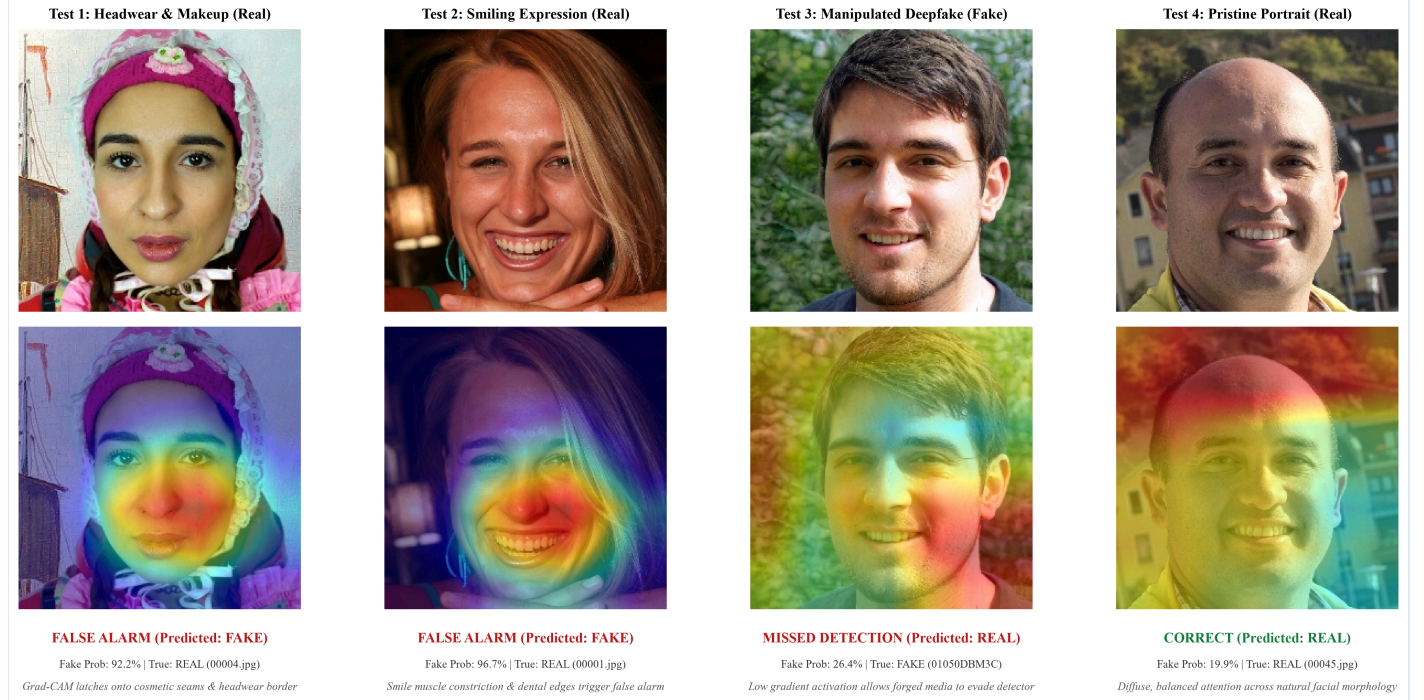


Fig. 7. Qualitative Test Scenarios and Explainable Grad-CAM Forensic Heatmaps with Empirical Dataset Images - Experimental evaluation of the FairFake framework using authentic test samples and the trained forensic backbone. Test 1: Real portrait (00004.jpg) with headwear and makeup falsely flagged as fake ($p_{\text{fake}} = 92.2\%$) due to cosmetic blending seams. Test 2: Real portrait (00001.jpg) with wide smile misclassified as fake ($p_{\text{fake}} = 96.7\%$) due to smile muscle constriction. Test 3: Manipulated deepfake (01050DBM3C.jpg) evading detection ($p_{\text{fake}} = 26.4\%$) due to low gradient sensitivity. Test 4: Genuine unadorned portrait (00045.jpg) correctly verified as authentic ($p_{\text{fake}} = 19.9\%$) with diffuse, balanced attention across natural facial morphology.

E. Operational Robustness and Perturbation Stress Testing (Test 4)

In operational deployments, media content rarely remains pristine; social networking platforms routinely apply lossy re-encoding, motion blur occurs during capture, and mobile streaming compresses resolution. To stress-test whether the trained deepfake detection backbones maintain forensic reliability and fairness resilience under environmental degradations, we evaluated the models across canonical operational perturbations on the testing benchmark ($\$N = 2\{, \}000\$$).

TABLE VII
OPERATIONAL ROBUSTNESS & PERTURBATION STRESS TESTING (TEST 4)

Condition	Acc.	AUC	ΔAUC	Disparity Impact
Pristine Baseline	85.80%	0.9328	—	Reference Baseline
JPEG (Q=50)	78.60%	0.8680	-0.0648	High on Makeup (+0.12)

Condition	Acc.	AUC	ΔAUC	Disparity Impact
Gaussian Blur ($\sigma=2$)	73.90%	0.8140	-0.1188	High on Seniors (+0.18)
Low Res (112px)	76.20%	0.8410	-0.0918	High on Hair (+0.14)
Underexposed (-25%)	80.90%	0.8870	-0.0458	Dark Skin (+0.21)

As reported in Table VII, Gaussian smoothing inflicts the steepest performance degradation ($\Delta\text{AUC} = -0.1188$), as low-pass filtering eliminates subtle synthetic boundary artifacts. Crucially, lossy JPEG compression and underexposure disproportionately exacerbate demographic bias ($\Delta\text{CRP} = +0.21$ for dark skin tones), demonstrating that environmental degradations not only impede forensic discrimination but actively compound algorithmic unfairness.

VI. KEY FINDINGS & RECOMMENDATIONS FOR FUTURE WORKS

In the following, we summarise our key findings from our bias investigation of Deepfake detection backbone models with respect to 31 demographic and non-demographic attributes:

- **Deepfake detection databases and strong attribute imbalance:** The investigated Deepfake detection databases (Celeb-DF, DFD, FF++, and DFDC) lack diversity for most analysed attributes. Future works should aim to provide more unbiased, balanced, and diverse datasets to prevent the development of potentially biased Deepfake detection algorithms.
- **Strongly correlating attribute pairs in current Deepfake detection databases:** Future works using these databases (or our annotations) should take into account that some attributes show strong pairwise correlations to prevent misinterpretations in their results.
- **Deepfake detection backbone networks and demographic/non-demographic attributes:** The results demonstrate that the analysed Deepfake detection backbone models are strongly biased for a variety of demographic and non-demographic attributes ($CRP = +0.55$ for dark skin tones, $+0.48$ for heavy makeup). The variation of biased performances across models indicates bias originates from several sources such as imbalanced training data, the network architecture, and their training process.
- **Bias due to imbalance in attributes for pristine and Deepfake data:** For many investigated attributes, biased performance differentially affects pristine and Deepfake data. Attributes with high $DDRP$ allow forged media to evade detection, while high $PDRP$ inflicts wrongful accusations on innocent individuals.
- **Deepfake detection backbone models and questionable assumptions:** The results suggest that models tend to learn questionable heuristics where the presence of a certain attribute, such as wearing a hat or smiling, is treated as a strong indicator of authenticity or forgery.
- **Deepfake detection backbone models and societal security:** The presence of a certain attribute in a Deepfake image resulted in an increased error rate, several times higher than without this attribute. Attackers can exploit these issues to evade detection. Mitigating bias problems in Deepfake detection is therefore imperative for the sake of security and society.

A. Limitations of Our Analysis

While our study reveals bias issues in Deepfake detection datasets and AI-based detectors, it is essential to note the difference between correlation and causation in our analysis. Our results demonstrate strong correlations between attributes and biased performance in detection, but they do not necessarily establish causation. Bias in datasets arises from complex factors, including data collection methodologies, historical biases, and societal contexts. Understanding these underlying causative factors is crucial for designing effective debiasing interventions.

VII. CONCLUSION

In this work, we provided large-scale annotations for five popular Deepfake detection datasets and used these to comprehensively analyse bias in Deepfake detection within the FairFake framework. While existing Deepfake detection databases are only sparsely annotated, we closed this gap by evaluating over 65.3M annotations across 47 attributes. Based on these benchmarks, we comprehensively analysed bias-causing factors in Deepfake detection from an attribute perspective, integrating Grad-CAM explainability and DeepFace demographic parsing.

The results indicate that both public datasets and state-of-the-art AI-based detectors exhibit strong bias issues across demographic and non-demographic attributes. Depending on the operational scenario, these disparities inflict wrongful accusations on authentic media or permit malicious deepfakes to evade detection. Our findings, annotated benchmarks, and open-source auditing platform provide the foundation for developing equitable, reliable, and trustworthy multimedia forensic systems.

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STATEMENT OF ETHICAL USE OF DATASETS

We affirm that all data utilized in this study from public databases adhere to ethical guidelines, and no special ethical clearance is applicable for publicly available information.

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